University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

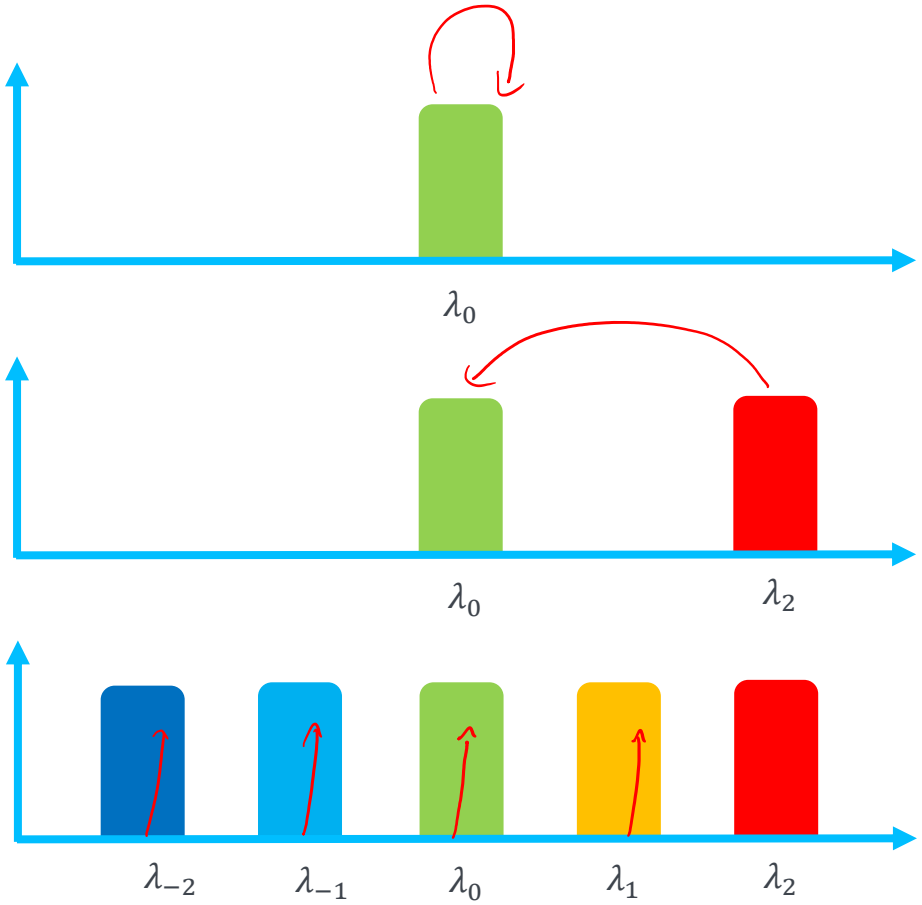
Lecture 14



Prof. Dr.-Ing. Georg Rademacher

Nonlinear inteference in WDM transmission

University of Stuttgart
Germany



Self-phase modulation

SPM

cross-phase modulation

XPM

Four-wave-Mixing

FWM

Four-wave mixing

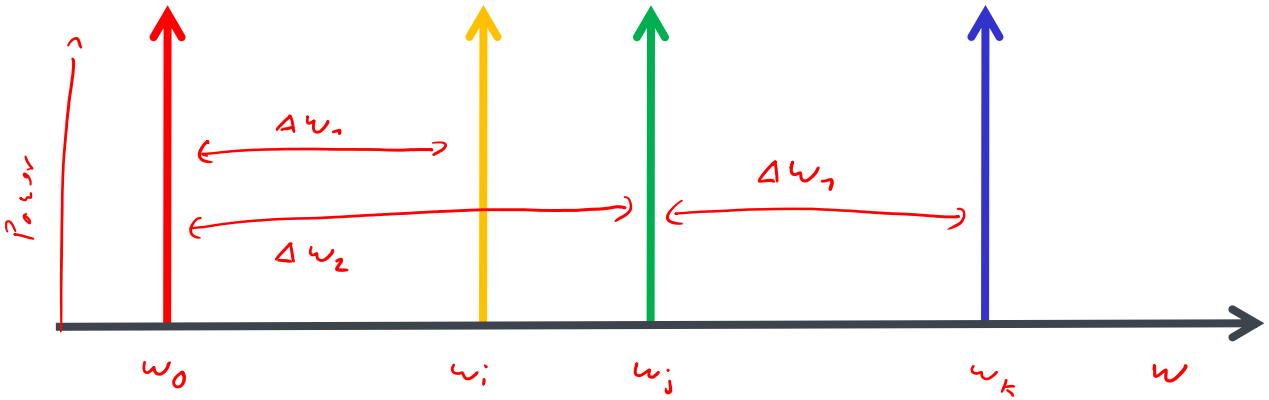
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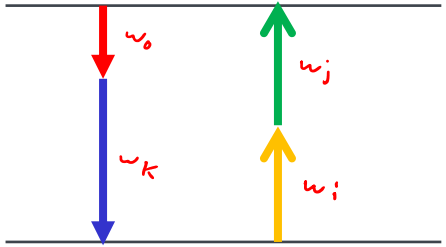
$\omega_i = \omega_0 + \Delta\omega_1$

$\omega_j = \omega_0 + \Delta\omega_2$

$\omega_k = \omega_0 + \Delta\omega_1 + \Delta\omega_2$

Conservation of energy





Four-wave mixing

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$$\omega_0 = \omega_i + \omega_j - \omega_k$$

frequency condition

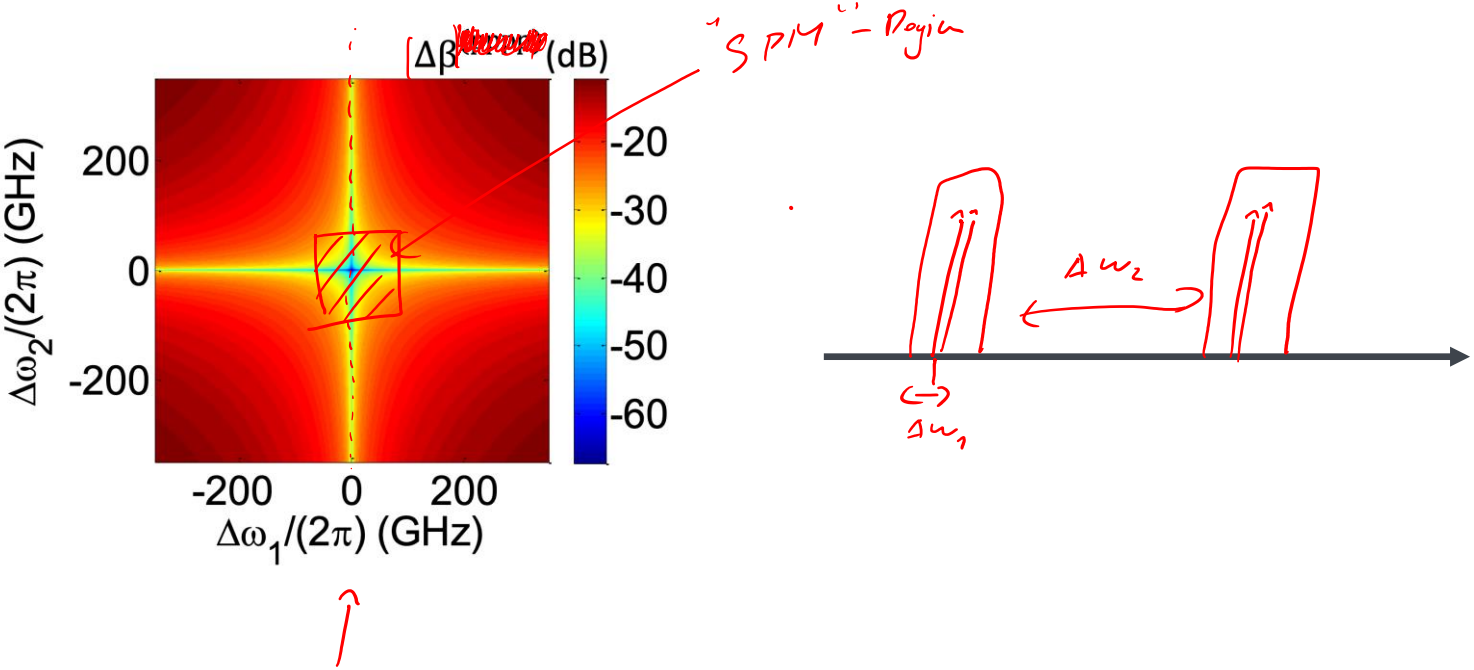
$$\Delta\beta = -\beta(\omega_0) + \beta(\omega_i) + \beta(\omega_j) - \beta(\omega_k)$$

phase-matching condition

$$\beta(\omega) = \beta_0 + \beta_1(\omega - \omega_0) + \frac{1}{2}\beta_2(\omega - \omega_0)^2 + \frac{1}{6}\beta_3(\omega - \omega_0)^3 + \dots$$

$$\Delta\beta = \beta_2(\Delta\omega_1 - \omega_0)(\Delta\omega_2 - \omega_0)$$

Four-wave mixing



Nonlinear optical signal-to-noise ratio

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Germany

$$OSNR_{NL} \quad \frac{P_{in}}{N\,(G_{ASE} + G_{NLI}) \cdot 12.5\,GHz}$$

$$G_{NLI} = (G_{Tx})^3 \,(\tilde{\gamma})^2 \iint_{B_{opt}} \eta(\Delta\beta)\, df_1\, df_2$$

$$\eta(\beta_2, \Delta f_1, \Delta f_2) = \left| \frac{1 - e^{-2\alpha L}\, e^{4\bar{\gamma}^2 j |\beta_2| (\Delta f_1 - f)(\Delta f_2 - f) L}}{2\alpha - 4\bar{\gamma}^2 |\beta_2| (\Delta f_1 - f)(\Delta f_2 - f)} \right|^2$$

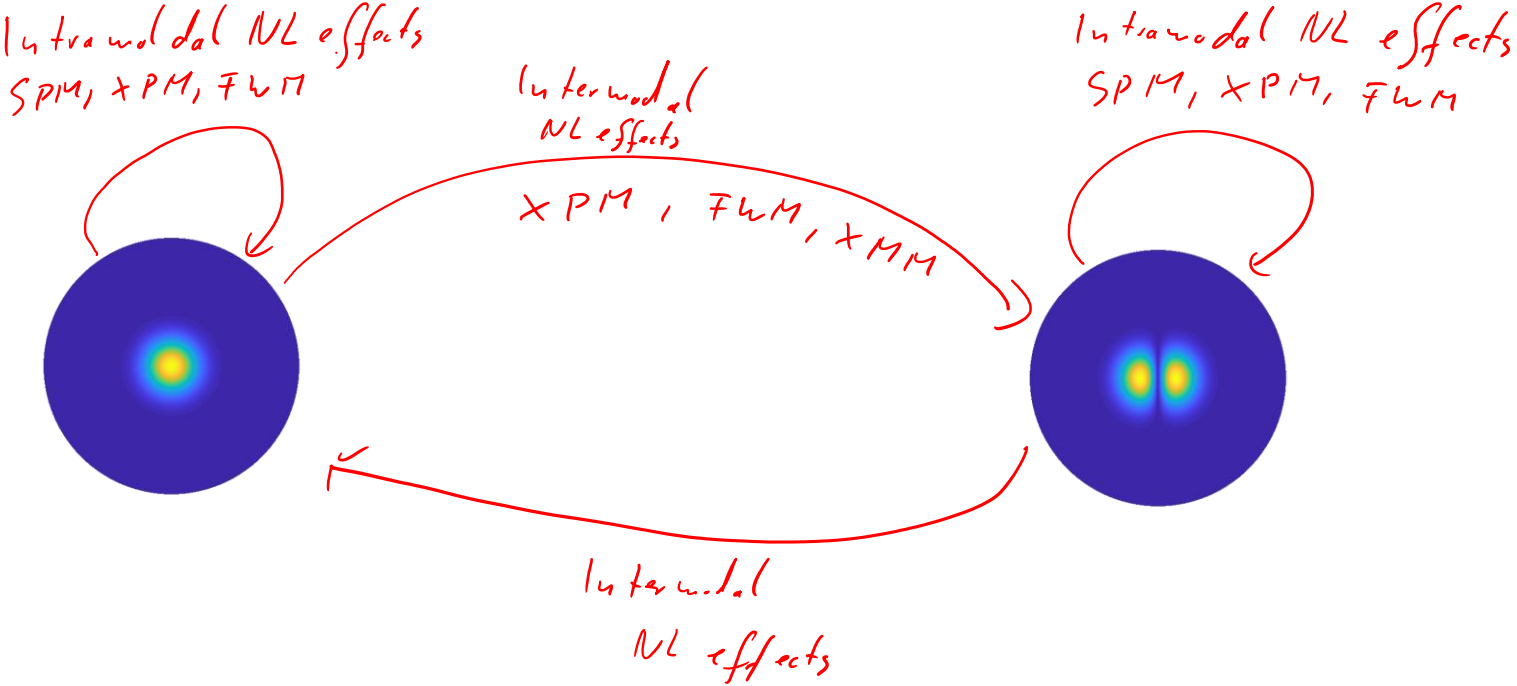
JOURNAL OF LIGHTWAVE TECHNOLOGY, VOL. 30, NO. 24, DECEMBER 15, 2012

The GN Model of Non-Linear Propagation in Uncompensated Coherent Optical Systems

Pierluigi Poggiolini, *Member, IEEE*

(Invited Paper)

Four-wave mixing



Intermodal four-wave mixing

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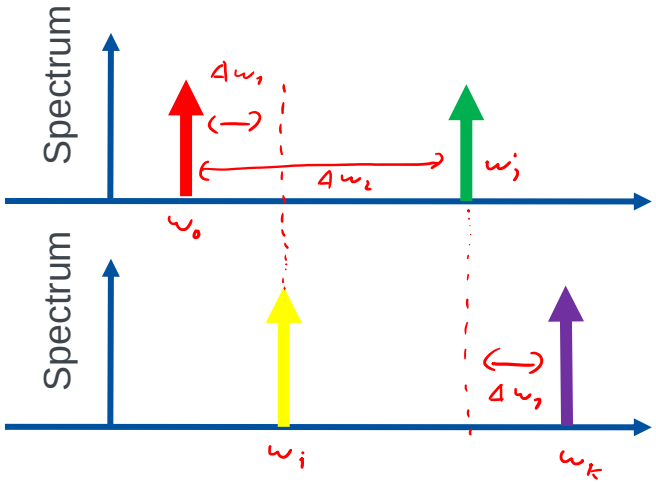
$$\Delta\beta^{(pqpq)}(\omega_0, \Delta\omega_1, \Delta\omega_2) = -\beta^{(p)}(\omega_0) + \beta^{(q)}(\omega_i) + \beta^{(p)}(\omega_j) - \beta^{(q)}(\omega_k)$$

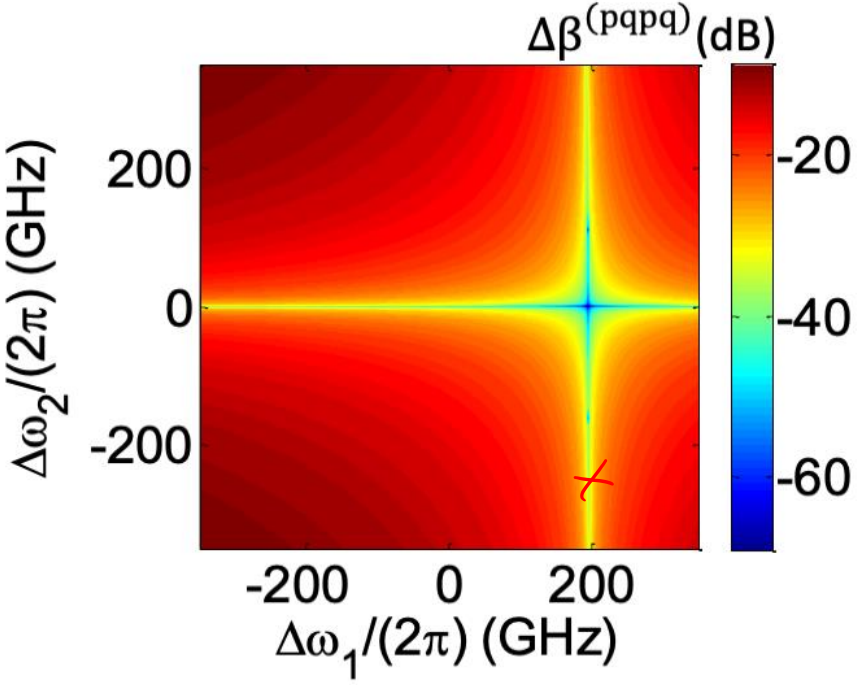
$$\beta^{(p)}(\omega) = \beta_0^{(p)} + \beta_1^{(p)}(\omega - \omega_0) + \frac{1}{2}\beta_2^{(p)}(\omega - \omega_0)^2 + \dots$$

$$\Delta\beta^{(pqpq)}(\Delta\omega_1, \Delta\omega_2) = \left(\beta_1^{(p)} - \beta_1^{(q)}\right)\Delta\omega_2 - \beta_2^{(p)}\Delta\omega_1\Delta\omega_2$$

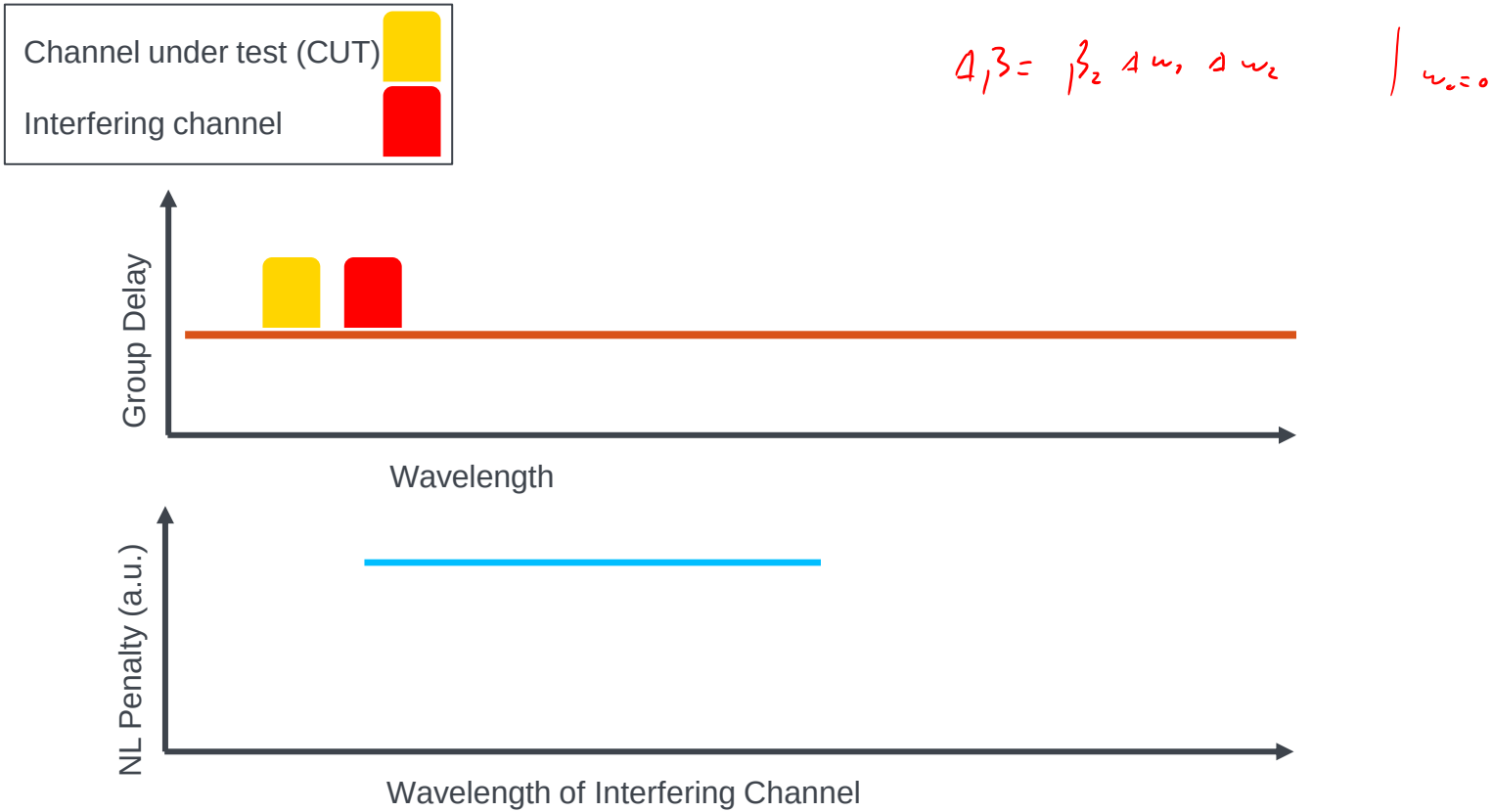
Intermodal four-wave mixing

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Germany

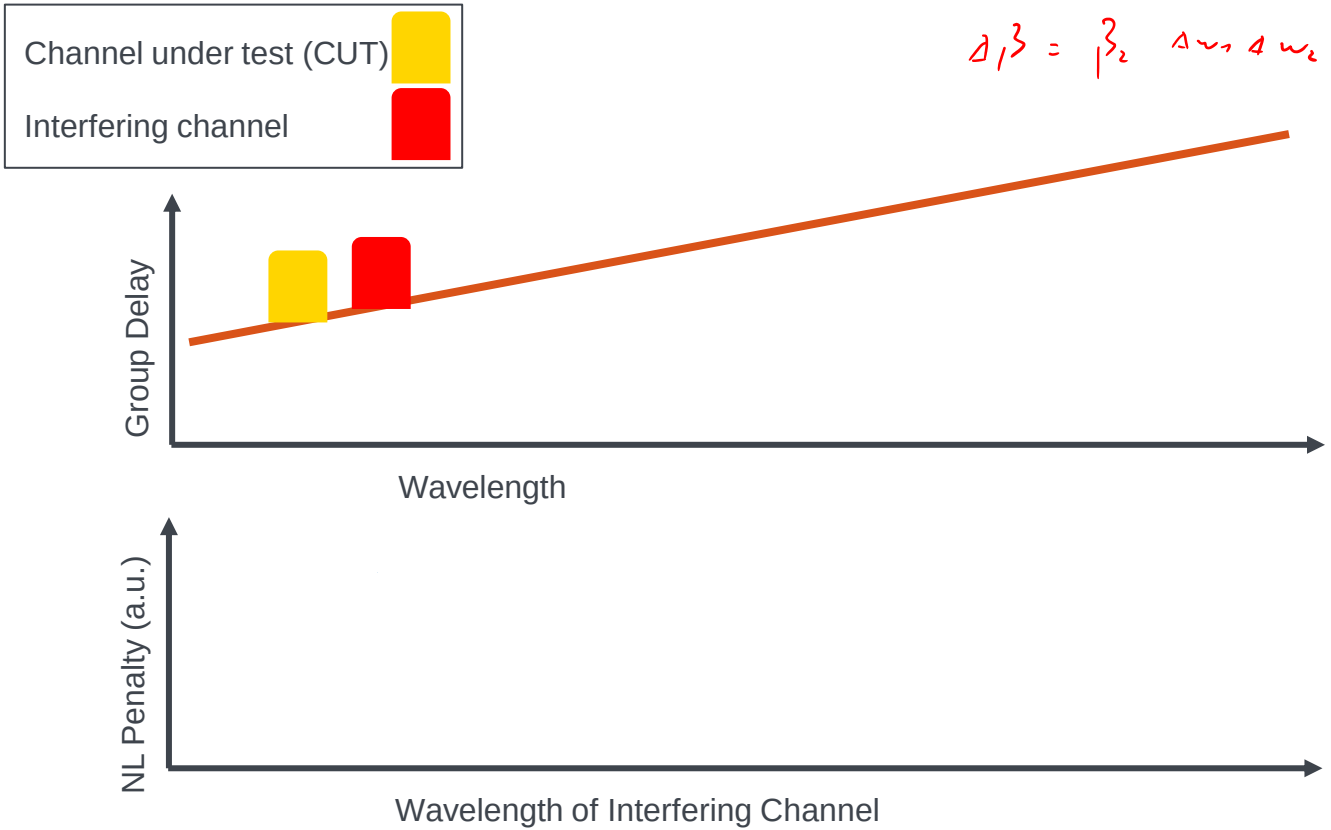




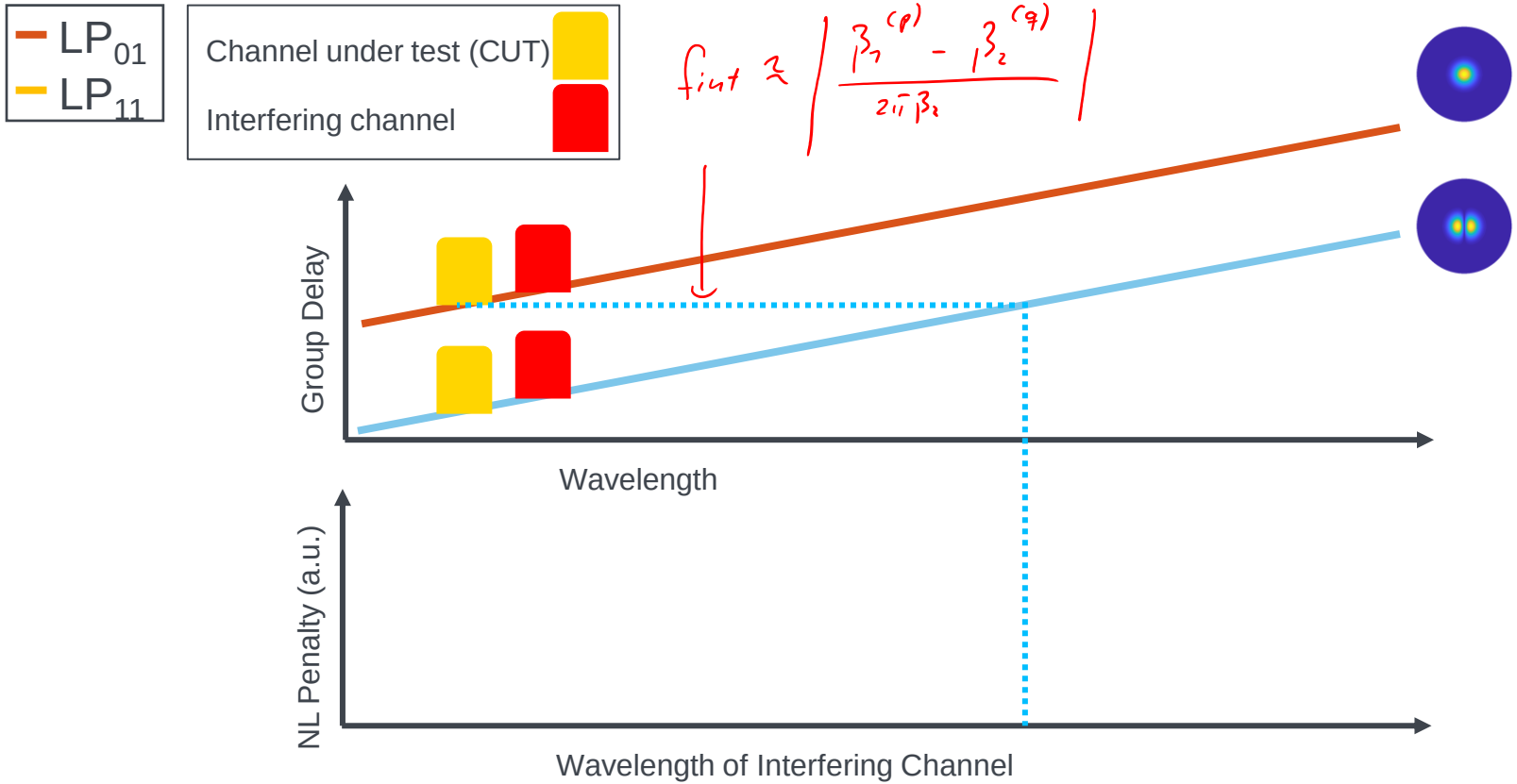
Nichtlineare Signalverzerrung in einer nicht dispersiven Single-Mode Faser



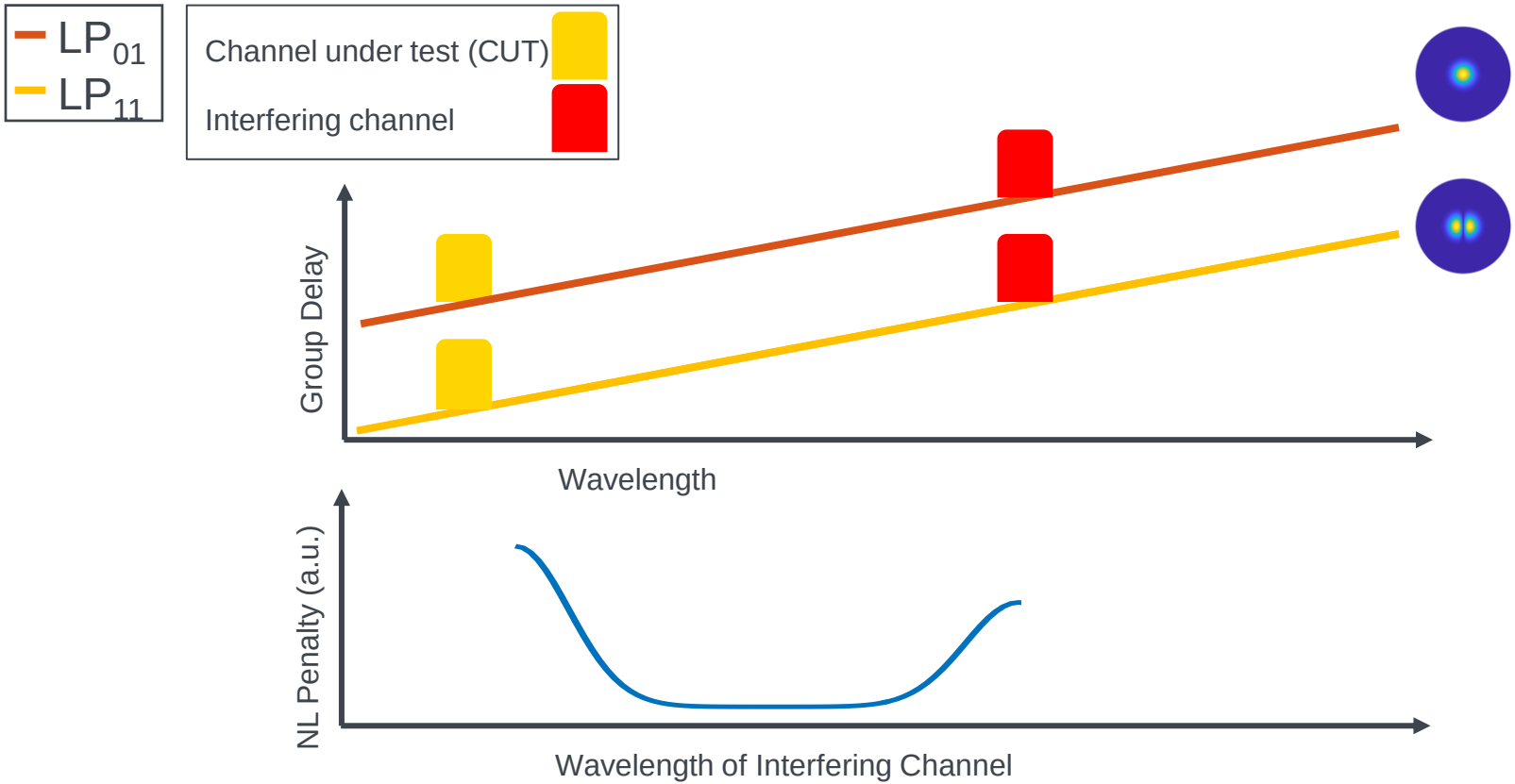
Nichtlineare Signalverzerrung in einer standard Single-Mode Faser



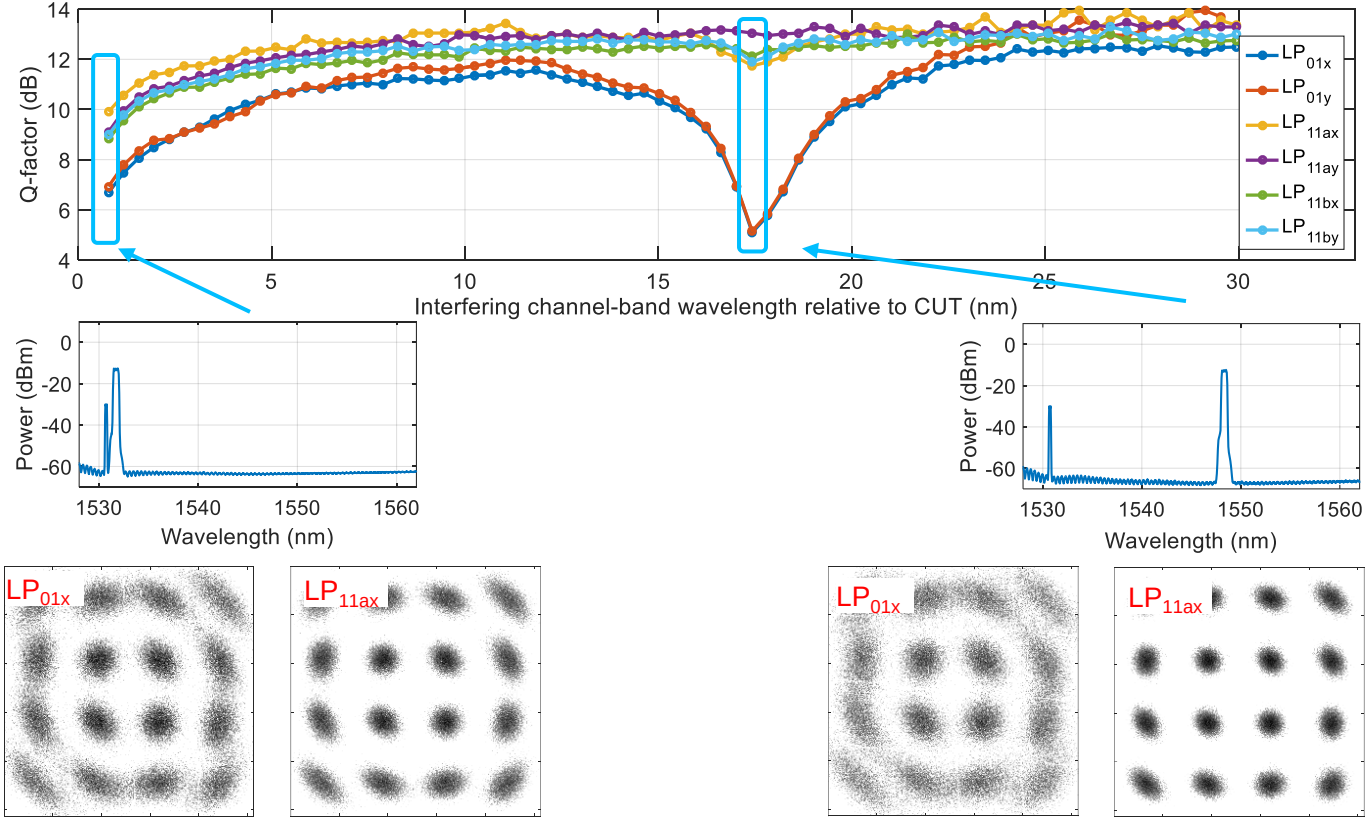
Nichtlineare Signalverzerrung in einer Few-Mode Faser



Nichtlineare Signalverzerrung in einer Few-Mode Faser

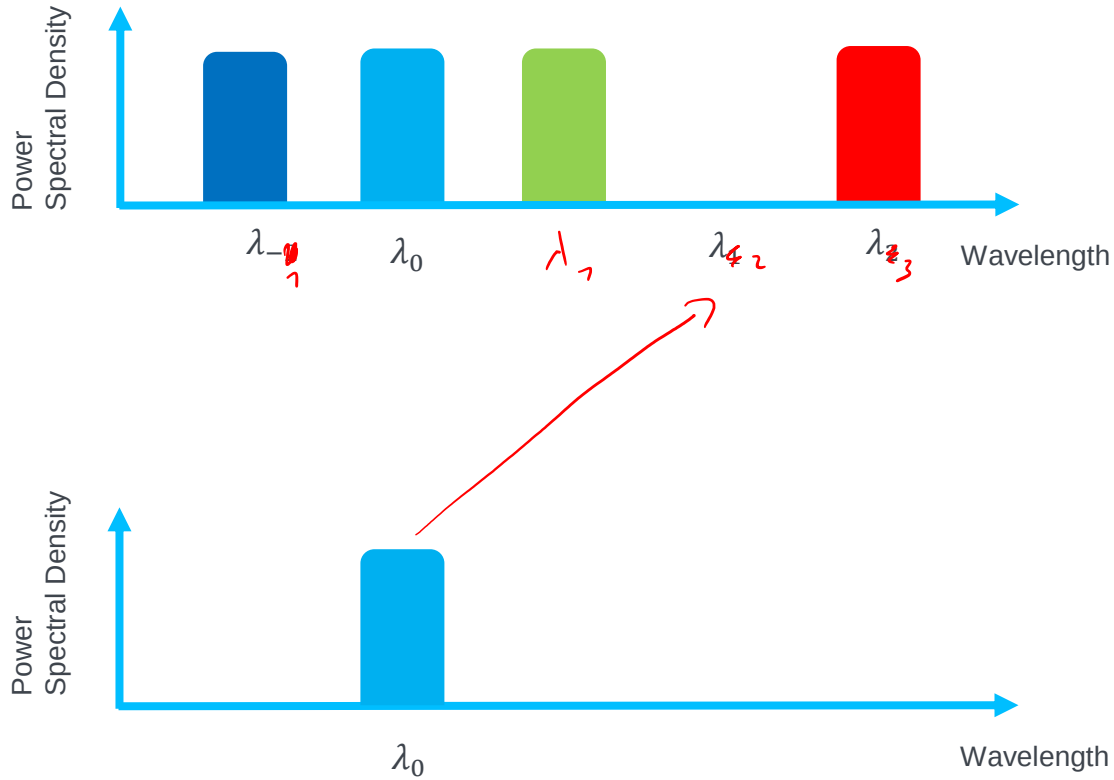


Experimentale Demonstration



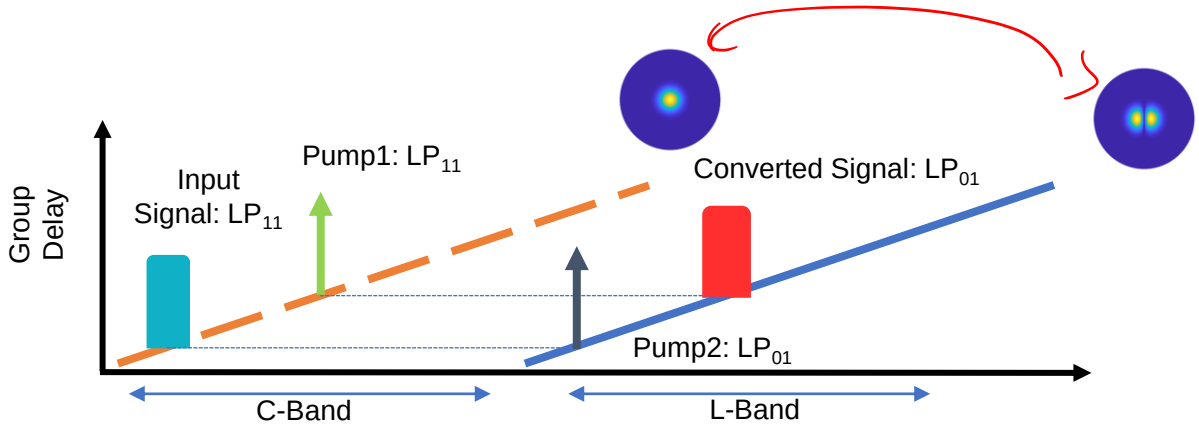
G. Rademacher et al., “Investigation of Intermodal Nonlinear Signal Distortions in Few-Mode Fiber Transmission”, Journal of Lightwave Technology vol. 37, 2019

Using intermodal FWM to perform all-optical signal processing

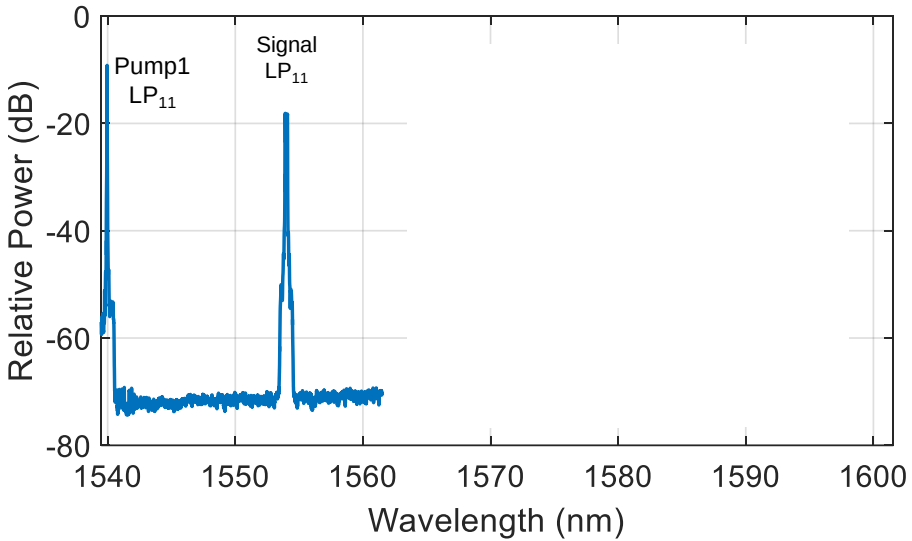


Investigated process: Phase Conjugation (PC)

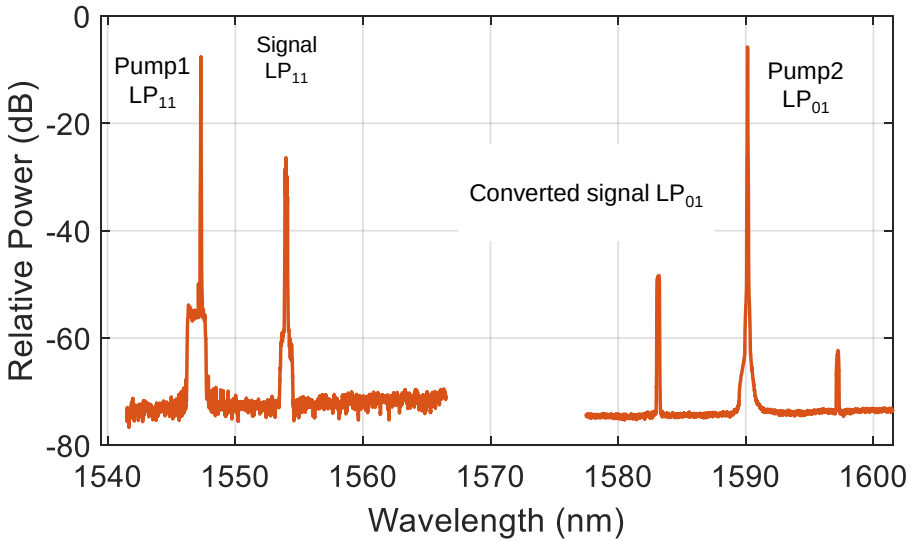
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Germany



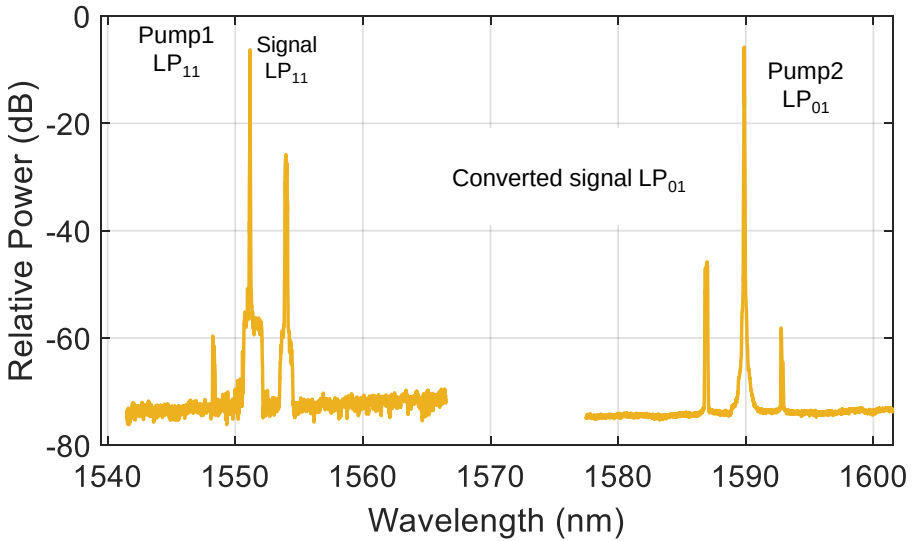
Results: Phase conjugation with single channel QPSK signals



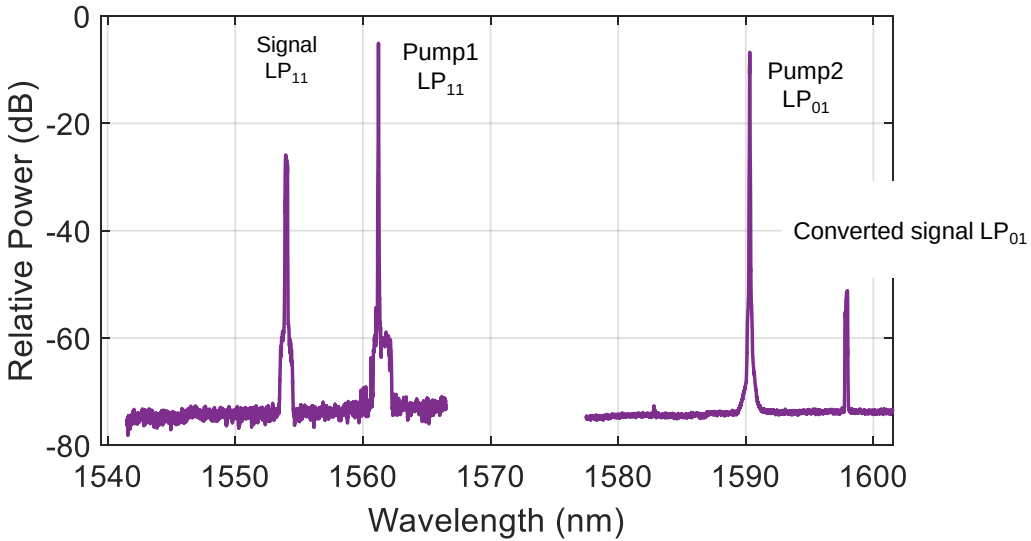
Results: Phase conjugation with single channel QPSK signals



Results: Phase conjugation with single channel QPSK signals

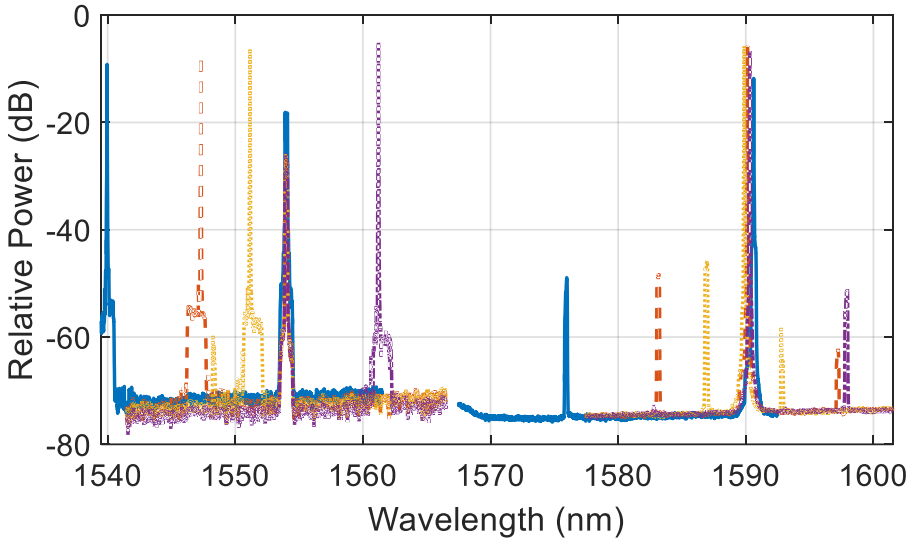


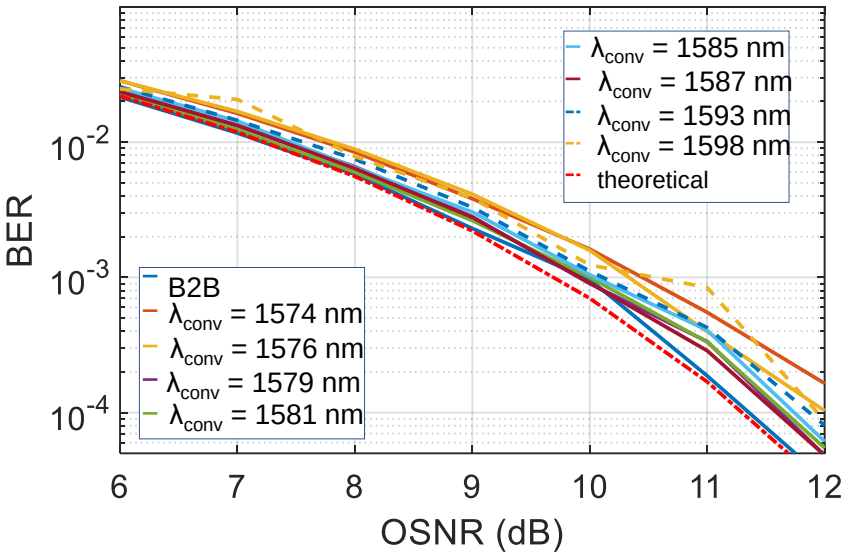
Results: Phase conjugation with single channel QPSK signals



Results: Phase conjugation with single channel QPSK signals

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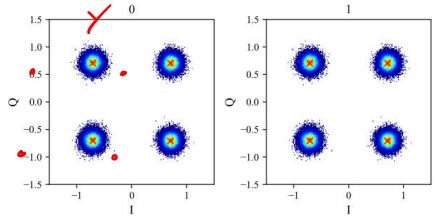


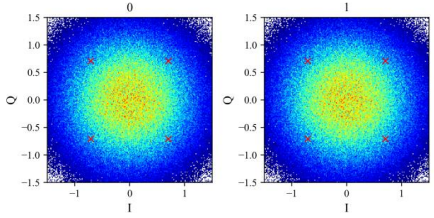


Principle of a coherent optical fiber transmission link

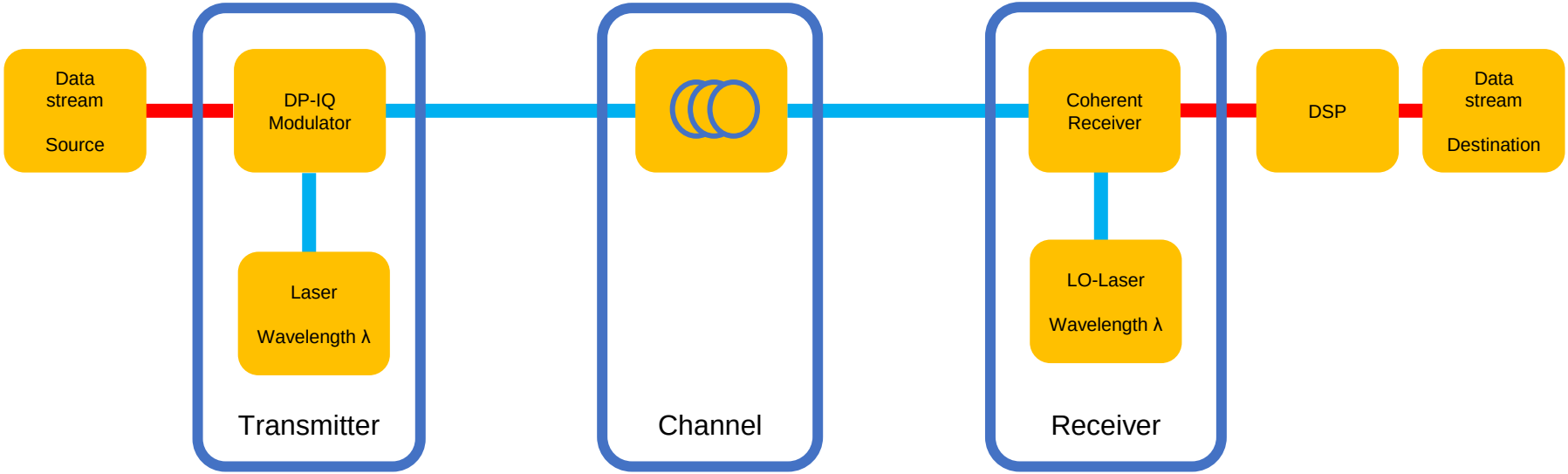
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Digital Signal Processing (DSP)

1164 IEEE JOURNAL OF SELECTED TOPICS IN QUANTUM ELECTRONICS, VOL. 16, NO. 5, SEPTEMBER/OCTOBER 2010

Digital Coherent Optical Receivers: Algorithms and Subsystems

Seb J. Savory, *Member, IEEE*

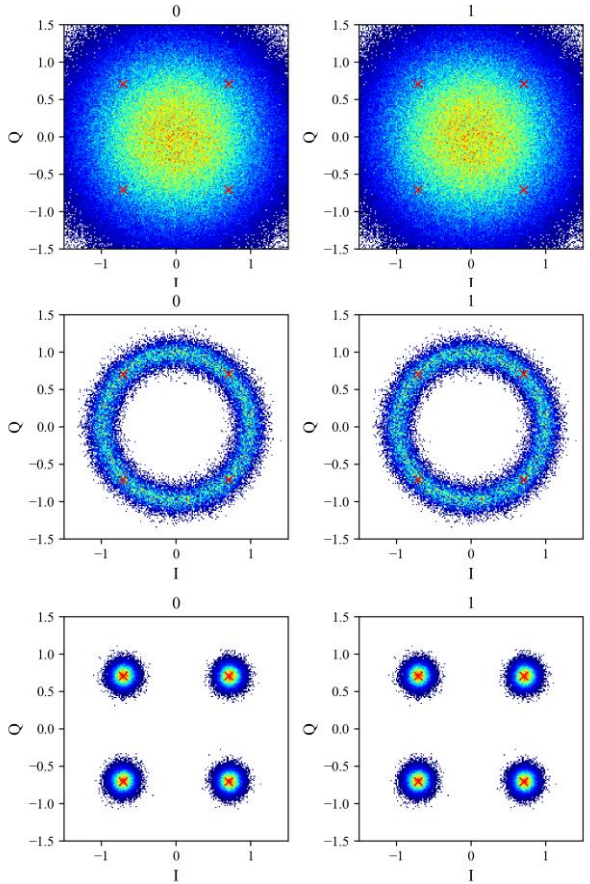
(Invited Paper)

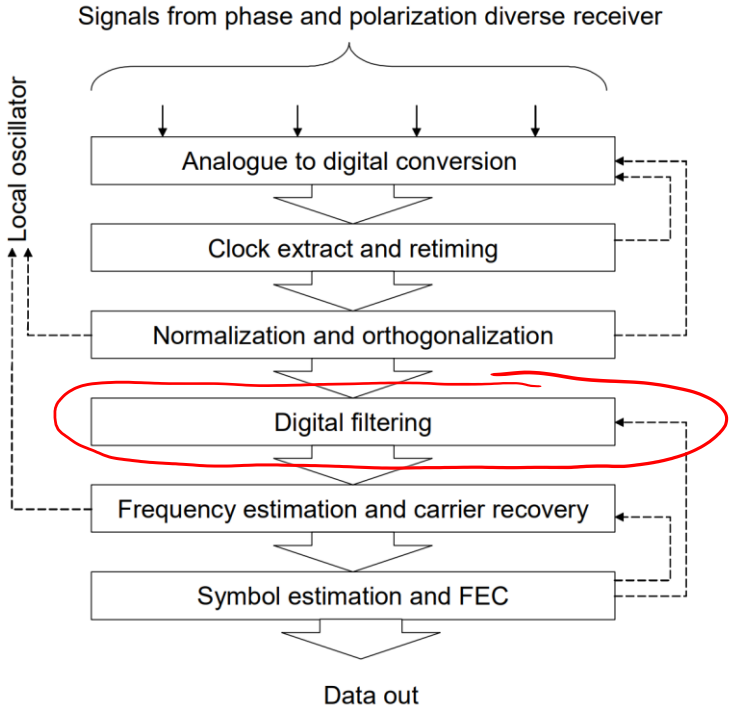
***Abstract*—Digital coherent receivers have caused a revolution in the design of optical transmission systems, due to the subsystems and algorithms embedded within such a receiver. After giving a high-level overview of the subsystems, the optical front end, the analog-to-digital converter (ADC) and the digital signal processing (DSP) algorithms, which relax the tolerances on these subsystems are discussed. Attention is then turned to the compensation of transmission impairments, both static and dynamic. The discussion of dynamic-channel equalization, which forms a significant part of the paper, includes a theoretical analysis of the dual-polarization constant modulus algorithm, where the control surfaces several different equalizer algorithms are derived, including the constant modulus, decision-directed, trained, and the radially directed equalizer for both polarization division multiplexed quadrature phase shift keyed (PDM-QPSK) and 16 level quadrature am-**

maximum data per wavelength (5.1 Tb/s) [9], and the maximum capacity distance product (112 Pbkm/s) [10], coherent detection also offers benefits at the network level. Not only can digital coherent receivers allow for more robust transmission, and hence, rerouting, but due to the presence of a local oscillator, the receiver is inherently frequency selective permitting new network architectures to be devised [11], [12].

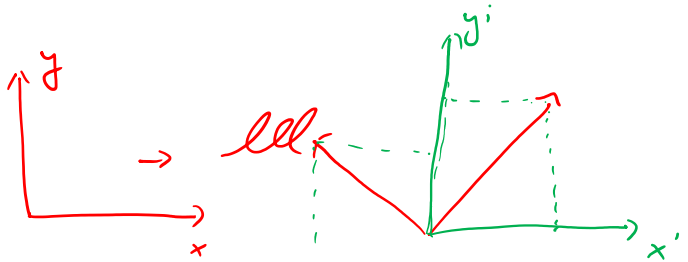
In order to understand the operation of the digital coherent receiver, in this paper, we consider each of the subsystems required. After giving a high-level overview of the subsystems in Section II, we discuss the optical front end, analog-to-digital converter (ADC) and the DSP algorithms necessary to overcome

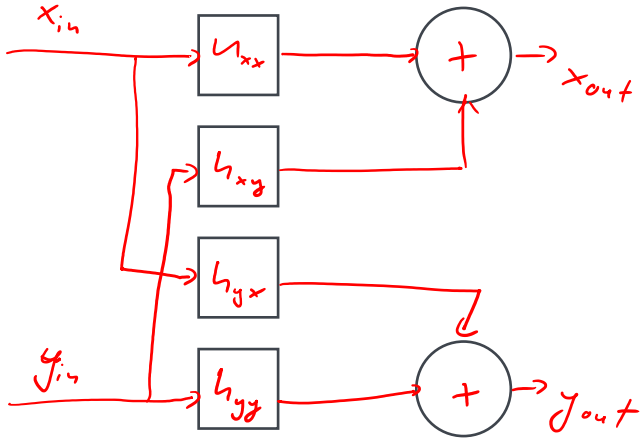
Digital Signal Processing (DSP)





Dynamic Channel Equalization





$$\begin{pmatrix} x_{out} \\ y_{out} \end{pmatrix} = \begin{pmatrix} h_{xx} & h_{xy} \\ h_{yx} & h_{yy} \end{pmatrix} \begin{pmatrix} x_{in} \\ y_{in} \end{pmatrix}$$

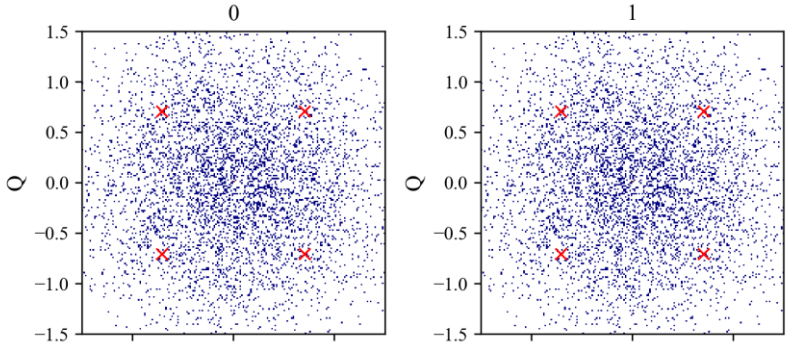
$$\varepsilon_x = 1 - |x_{out}|^2$$

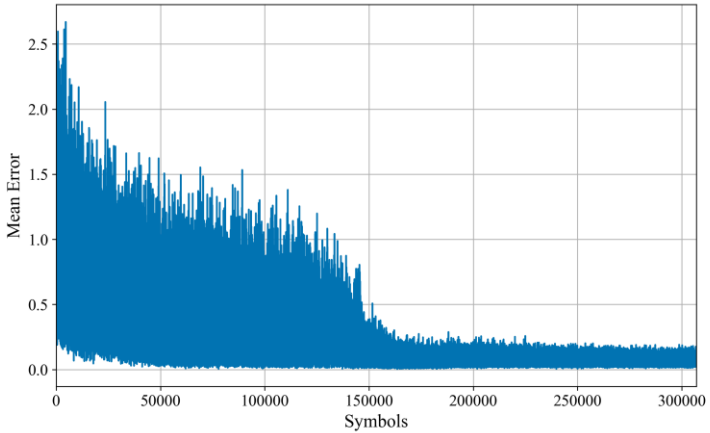
$$\varepsilon_y = 1 - |y_{out}|^2$$

$$h_{xx} \rightarrow h_{xx} - \frac{\mu}{2} \frac{d|\varepsilon_x|^2}{dh_{xx}}$$

How does this look in reality?

University of Stuttgart
Germany





662 JOURNAL OF LIGHTWAVE TECHNOLOGY, VOL. 28, NO. 4, FEBRUARY 15, 2010

Capacity Limits of Optical Fiber Networks

René-Jean Essiambre, *Senior Member, IEEE, Fellow, OSA*, Gerhard Kramer, *Fellow, IEEE*,
Peter J. Winzer, *Fellow, IEEE*, Gerard J. Foschini, *Fellow, IEEE*, and Bernhard Goebel, *Student Member, IEEE*

(Invited Paper)

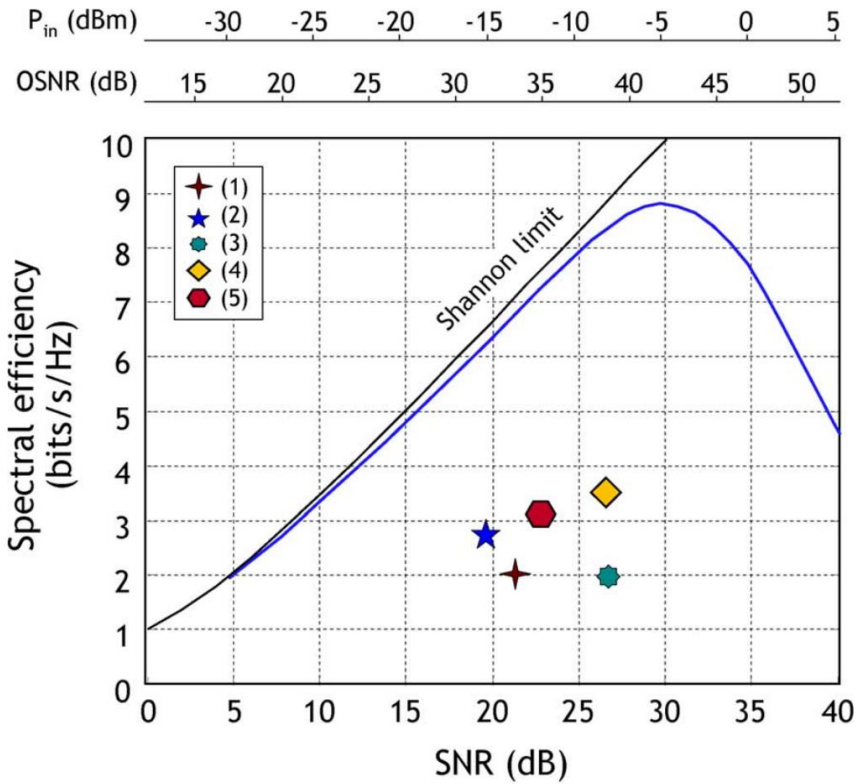
***Abstract*—We describe a method to estimate the capacity limit of fiber-optic communication systems (or "fiber channels") based on information theory. This paper is divided into two parts. Part 1 reviews fundamental concepts of digital communications and information theory. We treat digitization and modulation followed by information theory for channels both without and with memory. We provide explicit relationships between the commonly used signal-to-noise ratio and the optical signal-to-noise ratio. We further evaluate the performance of modulation constellations such as quadrature amplitude modulation, combinations of am-**

and practical interests [1]–[4]. Such a limit is referred to as the *channel capacity* and the process of evaluating this limit leads to a better understanding of the technologies needed to approach it.

Capacity evaluations require information theory [1] that must be adapted to the specific characteristics of the channel under study. A pragmatic approach is to define the channel as "that part of a communication system that the designer is unable or un-

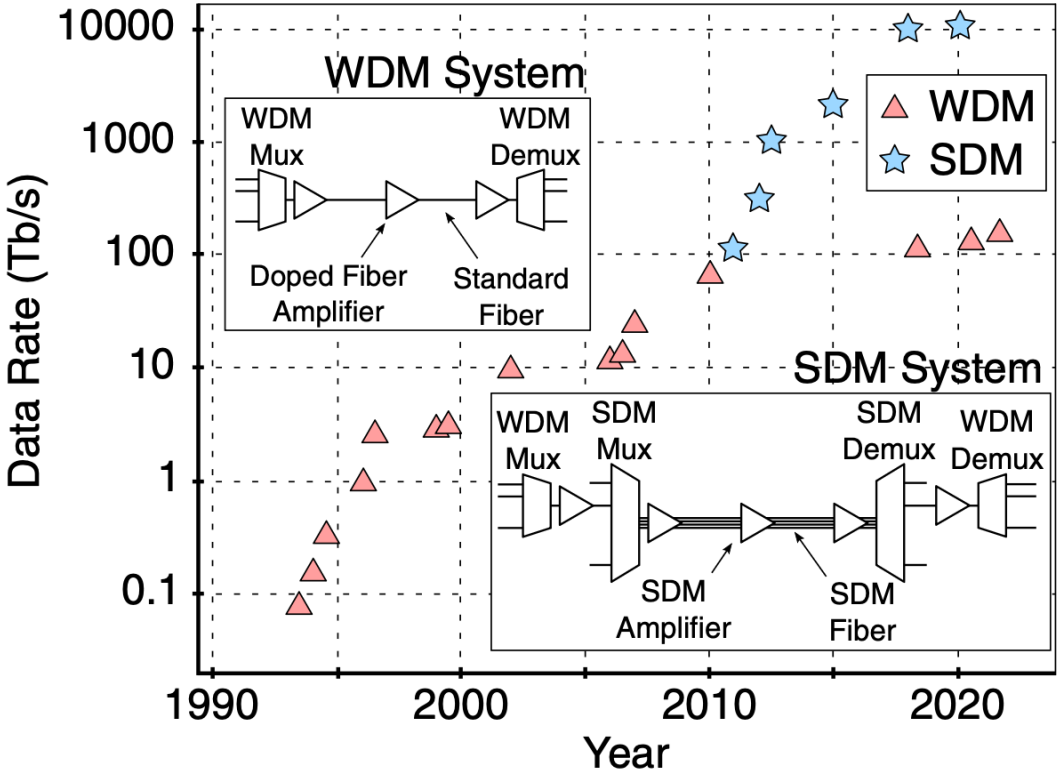
Capacity Limits of Optical Communications systems

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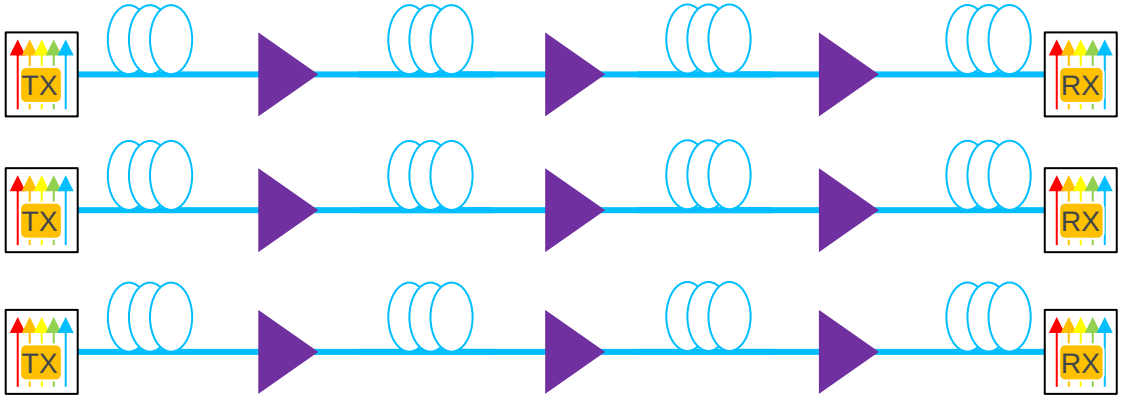


Space Division Multiplexed Transmission

University of Stuttgart
Germany

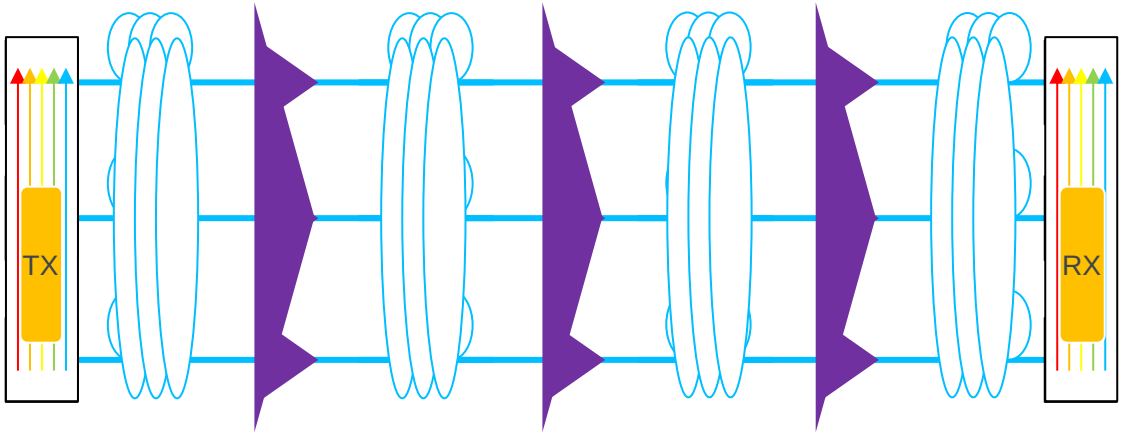


Space-division multiplexing (SDM)



Space-division multiplexing (SDM)

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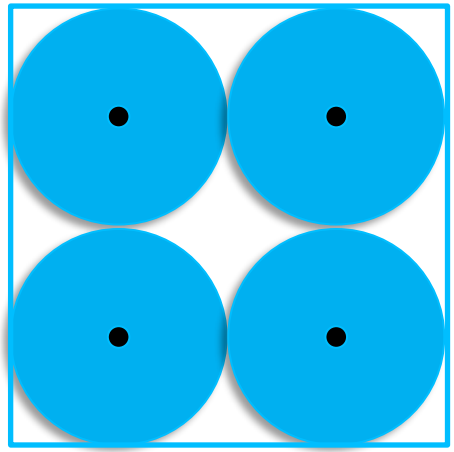


Optical fibers for space-division multiplexing

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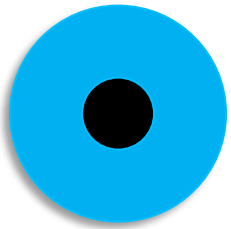
Fiber bundle

$\sim 125\ \mu m$



Few-/ multi-mode fiber

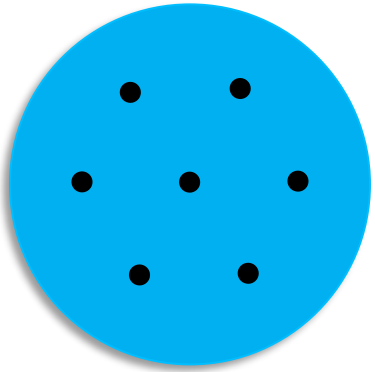
$\sim 125\ \mu m$



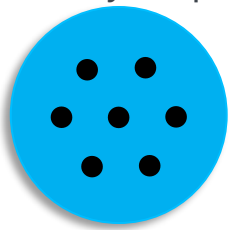
Multi-core fiber

$\sim 125\ -\ 250\ \mu m$

Weakly coupled



Randomly coupled



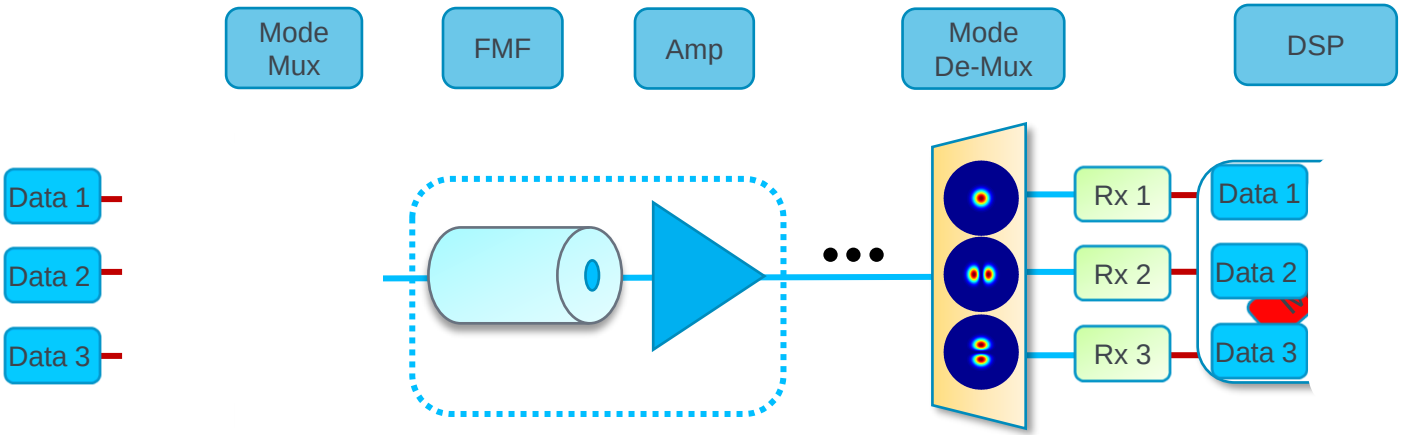
Few-mode-multi-core fiber

$\sim 160\ -\ 320\ \mu m$



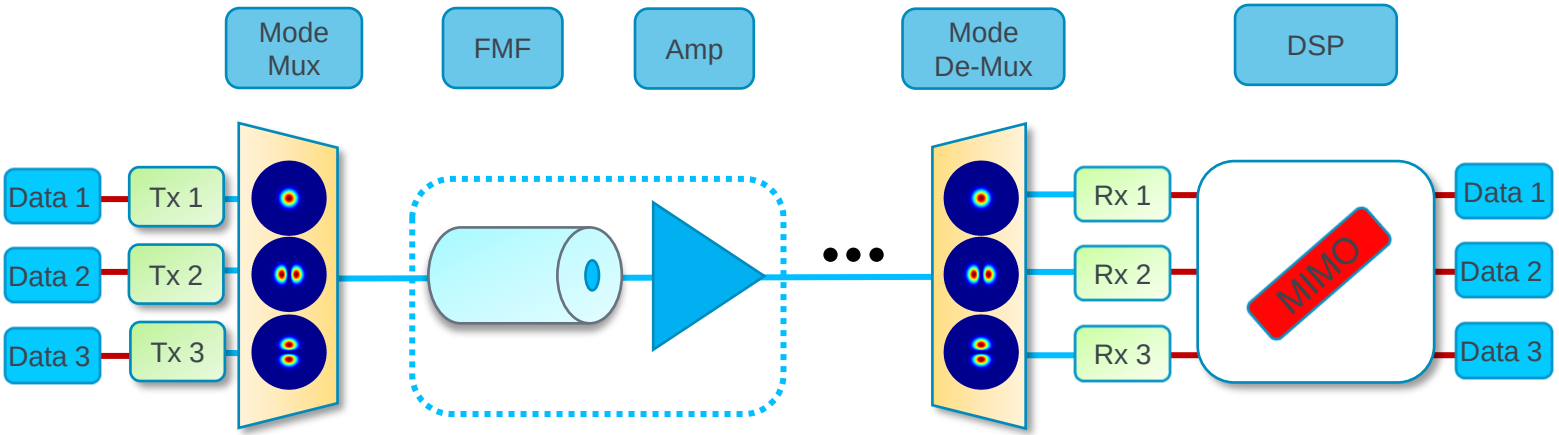
SDM transmission with a three-mode fiber

University of Stuttgart
Germany



SDM transmission with a three-mode fiber

University of Stuttgart
Germany



University of Stuttgart
Institute of electrical and optical communications

Thank you!

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www. int.uni-stuttgart.de/

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